Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Data Warehousing, Distributed Data Systems, ETL Pipeline Development, SQL Query Optimization, Dimensional Data Modeling, Star/Snowflake Schemas, Slowly Changing Dimensions (SCD Type 1/2), Reverse Logistics Infrastructure, Predictive Modeling, Automated Data Quality Auditing, Agile Methodologies, Software Engineering Best Practices, Capacity Planning

Programming Languages & Tools: AWS (S3, Redshift), Azure (Data Factory, Synapse, SQL), Snowflake, Apache Spark, PySpark, Python, SQL, PL/SQL, Power BI, Tableau, MicroStrategy, Airflow, Scikit-Learn, XGBoost, KIRO IDE

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Senior Data Engineer, Reverse Logistics & Scalable Data Infrastructure — **May 2022 – Present**

- **Designed and architected massively scalable** distributed data warehousing systems and dimensional data models (Star/Snowflake schemas, SCD Type 1/2) utilizing AWS (S3, Redshift) and Azure Synapse to support high-throughput Reverse Logistics pipelines.
- **Optimized complex high-performance SQL scripts** and stored procedures to process large-scale, complex datasets, cutting operational turnaround time by over 80% with an automated Data Quality Audit framework.
- **Engineered robust end-to-end ETL pipelines** leveraging Apache Spark, PySpark, and Azure Data Factory, guaranteeing 99.9% data availability SLAs and reliable supply chain data ingestion.
- **Partnered with software development teams** to implement automated testing frameworks and Agentic AI tools (KIRO IDE, prompt engineering), accelerating data delivery cycles by 40% while upholding software engineering best practices.
- **Developed predictive machine learning models** using XGBoost, Scikit-Learn, and SARIMA for advanced capacity planning and operational forecasting, boosting operational reliability by 20%.

Master of Science in Information Systems & Graduate Certificate in Data Science — **May 2022 – Present**

- **Completed rigorous quantitative coursework** maintaining a 3.95/4.0 GPA at the University of Cincinnati, focusing on advanced data warehousing, database design, and statistical data analysis.
- **Executed comprehensive academic projects** applying machine learning algorithms and distributed computing frameworks to solve complex business and supply chain forecasting problems.
- **Mastered core data engineering principles** including relational database design, query optimization, data governance, and scalable pipeline development.

Bachelor of Engineering in Petrochemical — **May 2022 – Present**

- **Built strong analytical and problem-solving foundations** through rigorous engineering curricula at Pune University MIT, focusing on complex systems optimization and process automation.
- **Applied mathematical modeling techniques** to evaluate physical system performance and design efficient operational workflows.
- **Cultivated a self-starter mindset** and agile adaptability to thrive in fast-paced, highly technical environments.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst — **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.

- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.